

MUHAMMED ARSHAQ

Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-stack software engineer with 5+ years of experience designing, building and shipping web, mobile and AI-enabled products for startups and SMEs across Sri Lanka, the GCC and international markets. Experienced across the full product lifecycle, from requirements and system architecture to real-time features, payment integration, IoT telemetry and computer vision, with a consistent record of taking products from concept to production. As a co-founder of a software agency, brings strong ownership, client communication and delivery discipline. BSc (Hons) in Software Engineering from SLIIT (Second Class Upper, W GPA 3.63) with applied computer-vision research and an IEEE conference publication.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, PHP, SQL, C, C++, HTML, CSS

Frameworks & Libraries: React, Next.js, React Native, Node.js, NestJS, Express.js, Redux, Zustand, Tailwind CSS

Architecture: Microservices, REST, gRPC, WebSockets (Socket.io), event-driven systems, API gateways

AI / ML: PyTorch, Ultralytics YOLOv8, ONNX Runtime, computer vision, model training & deployment

Databases & ORM: PostgreSQL, MongoDB, MySQL, SQLite, Prisma ORM

DevOps & Testing: Docker, Docker Compose, NX monorepo, Git/GitHub, Linux, Jest, Playwright, Grafana k6

Spoken Languages: English (professional), Sinhala, Tamil

PROFESSIONAL EXPERIENCE

FocalDive (Pvt) Ltd - Full-Stack Software Engineer & Co-Founder

Feb 2021 - Present

Software agency: web, mobile, booking, e-commerce, SaaS and AI solutions

Puttalam, Sri Lanka

- Design, build and ship full-stack web, mobile and AI products for startups and SMEs in Sri Lanka, the GCC and global markets, from architecture through deployment.
- Delivered production platforms with real-time tracking, payments and wallets, IoT fleet telemetry, booking engines and admin dashboards (see Selected Projects).
- Design backend services and APIs, including microservice and event-driven architectures, with a focus on reliability, maintainability and performance under load.
- Lead technical delivery in Agile teams, covering requirements, estimation, code reviews and releases across 15+ client projects, and mentor junior developers and interns.

IMARA Software Solutions

Nov 2021 - Nov 2024

Software development company building custom web and mobile applications for business clients

Puttalam, Sri Lanka

Web Application Developer (Full-time)

Jul 2022 - Nov 2024

- Developed and maintained production MERN-stack web applications, including CRM functionality, focusing on performance, maintainability and user experience.
- Integrated REST APIs to fetch and manipulate data, streamlining communication between frontend clients and backend services.

Mobile Application Developer - Intern (Full-time)

Nov 2021 - Jun 2022

- Built cross-platform mobile app features, translating Figma designs into production screens and integrating backend APIs.

RESEARCH

BrineX - Integrated Climate-Intelligent Salt Production Ecosystem

2025 - 2026

Final-Year Research Project, SLIIT (Ref: 25-26J-431) - four-member team

Puttalam, Sri Lanka

- Four-module AI platform (forecasting, planning, vision quality control, federated learning) built as NestJS microservices in an NX monorepo.

My component: Salt Crystal Impurity Detection and Quality Grading Using Real-Time Computer Vision

- Designed and delivered the platform's computer-vision quality-control module, the first AI-based visual inspection system for tropical salt production, detecting pure, impure and foreign-material classes.
- Built a custom dataset from field capture at Puttalam salterns: 650 annotated images (Label Studio, CVAT) expanded to 1,952 via augmentation; benchmarked MobileNetV2, ResNet-50 and YOLOv8 before selecting a detection approach.
- Achieved over 99% detection accuracy on unseen test images, correctly identifying 99.7% of impure salt samples - the class that matters most for product quality.

- Engineered a production NestJS/TypeScript microservice with ONNX Runtime inference, gRPC contracts, MongoDB persistence and Socket.io streaming to a Next.js 14 dashboard, holding under 50 ms per frame (5 FPS) on CPU-only hardware.

PUBLICATIONS

T. Ansar, I. Sirimanna, **M. Arshaq**, R. Perumbuli, N. Silva and T. Jayalath, "From Pond to Market: A Two-Stage AI Forecasting System for Salt Production Planning in Puttalam, Sri Lanka," in *Proc. 8th Int. Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA 2026)*, IEEE Türkiye Section, Ankara, Türkiye, May 2026 (oral presentation).
IEEE Xplore: ieeexplore.ieee.org/document/11537206

SELECTED PROJECTS

Molt - Real-Time Livestock Monitoring System (FocalDive) **Mar 2026 - Present**
DFCCNet density regression, IP cameras, SMS & in-app alerts

- Replaced twice-daily manual poultry headcounts with continuous IP-camera monitoring using DFCCNet density regression, automated threshold alerts and SMS/in-app notifications. Standardised 600px frames and introduced a single inference queue to eliminate out-of-memory failures as camera count increased.
- Chose density regression over object detection to count overlapping birds accurately, and removed out-of-memory failures by standardising frames to 600 px behind a single inference queue so memory stays flat as cameras scale.

PassQue - Patient Queue Management System (FocalDive) **Apr 2026 - Present**
React Native, Spring Boot *Role: Cross-Platform Mobile App Developer*

- Built the patient-facing React Native mobile app from scratch for a clinic queue system that digitises appointments and cuts physical waiting times.
- Delivered real-time queue position tracking, status notifications and an intuitive, accessible UI; integrated Spring Boot REST endpoints for queue, patient, doctor and clinic data.

Weels - End-to-End Food Ordering & Delivery Platform (FocalDive) **Oct 2025 - Jun 2026**
React Native, Nest.js, PostgreSQL

- Two connected cross-platform apps (customer and delivery partner) on a shared backend: restaurant browsing, cart, secure checkout, live order tracking, rider assignment and run history.
- Pushed every rider state change to the customer app in real time and moved dispatch from phone calls into an in-system queue; PostgreSQL relational constraints prevent partial or duplicate order assignment.

City Bikes & Segway - E-Scooter Rental & Tracking App, Qatar (FocalDive) **Jan 2025 - Feb 2026**
Cross-platform mobile, IoT fleet telemetry, in-app payments

- Self-service rental app for a Qatari micro-mobility operator covering live-map vehicle discovery, unlock, in-app payment and wallet-based 30-minute and 1-hour ride passes.
- Integrated IoT location telemetry into an admin dashboard for live fleet visibility, gated unlocks on payment authorisation, and built a bilingual English/Arabic UI with full RTL support.

BMP - Sports Arena Booking & Management Platform (FocalDive) **Apr 2025 - Nov 2025**
Next.js, Nest.js, PostgreSQL

- Unified players and turf owners on one calendar: location-based discovery with live availability and instant booking, plus an owner dashboard for slots, rates, discounts, payouts and month-on-month earnings.
- Prevented double bookings with a single source of truth and slot locking at confirmation; added recurring weekly slot configuration and a built-in wallet for balances, discounts and settlements.

WrapItUp - Gift Hamper Purchasing & Management Platform **Jan 2025 - May 2025**
Next.js, Full-Stack

- Built an online gift-hamper platform that replaces shop-to-shop buying. Users pick items, customise the box shape, contents and presentation, pay securely and track delivery, all in a few minutes.
- Designed curated entry points by relationship and occasion to narrow the catalogue quickly. Also integrated secure payment directly into the platform and built order tracking in from the start, to reduce checkout drop-off and remove follow-up calls.

EDUCATION

BSc (Hons) in Information Technology, Specialising in Software Engineering **2022 - 2026**
Sri Lanka Institute of Information Technology (SLIIT) *Malabe, Sri Lanka*

- Second Class Upper Division (WGPA 3.63 / 4.00); Dean's List x4; SLIIT Scholarship Award for academic performance

G.C.E. Advanced Level - Physical Science (Physics, Chemistry, Combined Mathematics) **2018 - 2020**
President's Science College *Puttalam, Sri Lanka*